Causal Inference

Session 4 — Instrumental Variables

Alassane Diallo

Albert School · Master 2

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ALBERT SCHOOL

Where we left difference-in-differences

Big Idea 1 — When we cannot control who is treated

Difference-in-differences built the counterfactual from the control group's change over time. It needed **parallel trends**.

When the hidden factor that drives treatment also changes the trend, that assumption fails. We need another kind of evidence.

Remember noncompliance in randomized experiments: the lottery decided who was **offered** tutoring, but students decided whether to **attend**.

The offer was random. Attendance was not. Can the random offer tell us the effect of attending?

The tutoring example, with voluntary attendance

Big Idea 1 — When we cannot control who is treated

1,000 students. A lottery **offers** tutoring to 500 of them ($Z = 1$); the other 500 are not offered ($Z = 0$). Attending ($T = 1$) is each student's choice; a few non-offered students find tutoring on their own.

	offered $Z = 1$	not offered $Z = 0$
attend tutoring	60%	10%
average score	72	69

The **ITT** is $72 - 69 = 3$ points: the effect of *offering* tutoring (previous chapters). But the school asks another question:

By how much does *attending* tutoring raise a student's score?

Two tempting comparisons

Big Idea 2 — Comparing by attendance brings selection back

As treated

Attendees versus non-attendees, whatever the offer:

$$78.3 - 66.3 = 12.0 \text{ points.}$$

Per protocol

Offered students who attended versus the non-offered group:

$$77.3 - 69 = 8.3 \text{ points.}$$

Both compare groups that **chose** to attend with groups that did not. The motivated students who attend would have scored higher anyway: the selection bias of the first chapter is back.

The lottery protected the comparison **by offer**, not the comparison **by attendance**.

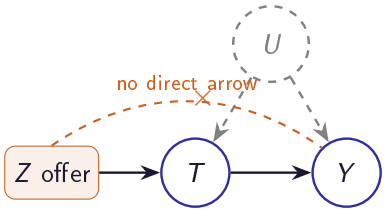
Can we use the random offer itself to learn the effect of attending?

The instrument

Big Idea 3 — An instrument: a push that is as good as random

The offer Z is an **instrument**: a variable that pushes some students into treatment, for reasons unrelated to their scores.

Motivation U still drives both attendance and scores, and it is not measured. But the offer reaches the score **only through attendance**, and the offer itself was drawn by lot.



Three conditions make Z a valid instrument:

1. **Relevance:** Z changes T. (The offer raises attendance from 10% to 60%.)
2. **Independence:** Z is as good as random. (It was drawn by lottery.)
3. **Exclusion:** Z affects Y only through T. (Receiving the offer letter does not raise scores by itself.)

What each condition buys

Big Idea 3 — An instrument: a push that is as good as random

Condition	In the tutoring example	Can we check it?
Relevance	the offer changes attendance	yes: compare attendance by offer
Independence	offer drawn by lottery	by design here; argued otherwise
Exclusion	the offer acts only by making students attend	never directly: it is an assumption

Relevance is in the data. Independence comes from the design. Exclusion must be argued from knowledge of the setting.

If these hold, how do we turn the offer's effect into the effect of attending?

Divide the effect of the offer by its effect on attendance

Big Idea 4 — The Wald estimator: rescale the ITT

The offer raised scores by 3 points on average (**reduced form**, the ITT).
The offer raised attendance by $60\% - 10\% = 0.5$ (**first stage**).

If the offer works only through attendance, the 3 points come entirely from the **half of the students** whose attendance it changed. Each of them must therefore have gained

$$\widehat{\beta}_{IV} = \frac{E[Y \mid Z=1] - E[Y \mid Z=0]}{E[T \mid Z=1] - E[T \mid Z=0]} = \frac{72-69}{0.60-0.10} = \frac{3}{0.5} = 6 \text{ points}$$

This is the **Wald estimator**. Per 100 offered students: 50 more attend, and the total score rises by $100 \times 3 = 300$ points, i.e. $300/50 = 6$ per student who switched.

Proof: the IV formula in a linear model

Big Idea 4 — The Wald estimator: rescale the ITT

MATHEMATICS

Suppose $Y = \alpha + \beta T + \varepsilon$, where ε contains motivation U, so $\mathrm{Cov}(T, \varepsilon) \neq 0$.

OLS is biased.

$$\frac{\mathrm{Cov}(Y, T)}{\mathrm{Var}(T)} = \beta + \frac{\mathrm{Cov}(T, \varepsilon)}{\mathrm{Var}(T)} \neq \beta.$$

Motivated students attend more and score more: $\mathrm{Cov}(T, \varepsilon) > 0$, upward bias.

IV is not. Take covariances with Z:

$$\mathrm{Cov}(Y, Z) = \beta\,\mathrm{Cov}(T, Z) + \underbrace{\mathrm{Cov}(\varepsilon, Z)}_{=0 \text{ (indep. + exclusion)}}$$

so, if $\mathrm{Cov}(T, Z) \neq 0$ (relevance),

$$\beta = \frac{\mathrm{Cov}(Y, Z)}{\mathrm{Cov}(T, Z)}$$

Binary Z gives Wald. With $p = P(Z = 1)$, $\mathrm{Cov}(Y, Z) = p(1-p)\big(E[Y \mid Z = 1] - E[Y \mid Z = 0]\big)$, and likewise for T. The $p(1-p)$ cancel: the ratio of covariances is the Wald ratio.

But whose effect is 6?

Big Idea 4 — The Wald estimator: rescale the ITT

The linear model assumed that tutoring has the **same effect for everyone**. The first chapter taught us that effects differ: struggling and strong students may gain differently.

And the offer does not move every student: some would attend anyway, some would never attend.

When effects differ, which students' effect does the Wald ratio measure?

Four kinds of students

Big Idea 5 — Whose effect? Compliers and the LATE

Each student has two potential attendance decisions: $T(1)$ if offered, $T(0)$ if not.

Type	$T(0)$	$T(1)$	In words
Always-takers	1	1	attend whatever the offer
Never-takers	0	0	never attend
Compliers	0	1	attend *because* they are offered
Defiers	1	0	attend *only* if not offered

We never see a student's type: we see only one of $T(1)$, $T(0)$. But we can count the groups. Assume no defiers (**monotonicity**: the offer never discourages anyone). Then

$$\underbrace{10\%}_{\text{attend, not offered}} = \text{always-takers}, \qquad 60\% - 10\% = \underbrace{50\%}_{\text{compliers}}, \qquad \underbrace{40\%}_{\text{rest}} = \text{never-takers}.$$

Example: the class, by type

Big Idea 5 — Whose effect? Compliers and the LATE MATHEMATICS

Suppose, for teaching, we could see each type's scores ($Y(0)$ without tutoring, gain from attending):

Type	share	$Y(0)$	gain
always-takers	10%	80	4
compliers	50%	70	6
never-takers	40%	64	?

Offered: $0.1 \times 84 + 0.5 \times 76 + 0.4 \times 64 = 72$.
Not offered:
$0.1 \times 84 + 0.5 \times 70 + 0.4 \times 64 = 69$.

Only the compliers' row differs between the two lines.

ITT $= 72 - 69 = 3 = 0.5 \times 6$ Wald $= 3/0.5 = 6 =$ **the compliers' gain**

As treated: $78.3 - 66.3 = 12.0$ Per protocol: $77.3 - 69 = 8.3$

Always-takers are stronger ($Y(0) = 80$), never-takers weaker (64): that is why comparing by attendance overstates the effect. The never-takers' gain is never revealed.

Proof: the LATE theorem (Imbens and Angrist, 1994)

Big Idea 5 — Whose effect? Compliers and the LATE MATHEMATICS

Assumptions. Independence: $Z \perp \{Y(t), T(z)\}$. Exclusion: Y depends on Z only through T. Relevance: $E[T(1) - T(0)] \neq 0$. Monotonicity: $T(1) \geq T(0)$ for everyone.

Reduced form. By independence, $E[Y \mid Z = 1] - E[Y \mid Z = 0] = E[Y(T(1)) - Y(T(0))]$. Split by type:

- always-takers and never-takers: $T(1) = T(0)$, so $Y(T(1)) - Y(T(0)) = 0$ (exclusion);
- defiers: none (monotonicity);
- compliers: $T(1) = 1,\ T(0) = 0$, so the difference is $Y(1) - Y(0)$.

$$E[Y \mid Z = 1] - E[Y \mid Z = 0] = P(\text{complier}) \cdot E[Y(1) - Y(0) \mid \text{complier}].$$

First stage. In the same way, $E[T \mid Z = 1] - E[T \mid Z = 0] = E[T(1) - T(0)] = P(\text{complier})$. Dividing:

$$\frac{E[Y \mid Z = 1] - E[Y \mid Z = 0]}{E[T \mid Z = 1] - E[T \mid Z = 0]} = E\big[Y(1) - Y(0) \mid \text{complier}\big] = \text{LATE} \qquad \blacksquare$$

The LATE is not the ATE

Big Idea 5 — Whose effect? Compliers and the LATE

The instrument identifies the effect for the students it **moves**: the **Local Average Treatment Effect** (LATE), local to the compliers.

What we learn: attending raises the score by 6 points for students who attend *because* they are offered.

What we do not learn: the gain of always-takers (here 4) or of never-takers (unknown), hence not the ATE.

A different instrument (a reminder text message, free transport) moves different students, and can identify a different LATE.

An instrument answers a question about the students it moves.

Next: how do we compute this on real data, with covariates and standard errors?

Two-stage least squares (2SLS)

Big Idea 6 — Two-stage least squares

Stage 1. Predict attendance from the offer:

$$T_i = \pi_0 + \pi_1 Z_i + v_i, \qquad \hat{\pi}_1 = 0.5.$$

Why it works: $\hat{T}$ keeps only the part of attendance that comes from the lottery. The part chosen by motivated students is left in v_i.

Stage 2. Regress the score on *predicted* attendance:

$$Y_i = \beta_0 + \beta_1 \hat{T}_i + e_i, \qquad \hat{\beta}_1 = 6.$$

Covariates (previous score, scholarship) can be added to both stages: they do not change the identification, they improve **precision**, as in randomized experiments.

Standard errors. Use the 2SLS formula from your software, not the standard errors of a hand-made second-stage OLS: those ignore that $\hat{T}$ was estimated.

Proof: 2SLS with one binary instrument is the Wald ratio

Big Idea 6 — Two-stage least squares MATHEMATICS

Stage 1 fits two values: $\hat{T} = \bar{T}_0$ when $Z = 0$ and $\hat{T} = \bar{T}_1$ when $Z = 1$, with $\bar{T}_z$ the attendance rate in arm z (0.10 and 0.60).

Stage 2 regresses Y on a variable that takes only these two values. The OLS slope through two group means is

$$\hat{\beta}_1 = \frac{\bar{Y}_1 - \bar{Y}_0}{\bar{T}_1 - \bar{T}_0} = \frac{72 - 69}{0.60 - 0.10} = 6,$$

exactly the Wald ratio of Big Idea 4.

Uncertainty. By the delta method, when the first stage is precisely estimated,

$$SE(\hat{\beta}_{IV}) \approx \frac{SE(\text{ITT})}{|\text{first stage}|}$$

With $SE(\text{ITT}) = 0.7$: $SE \approx 0.7/0.5 = 1.4$, and a 95% interval $6 \pm 1.96 \times 1.4 = [3.3,\ 8.7]$.

The denominator matters

Big Idea 6 — Two-stage least squares

The estimate is a ratio. Its precision depends on the denominator: how strongly the instrument moves treatment.

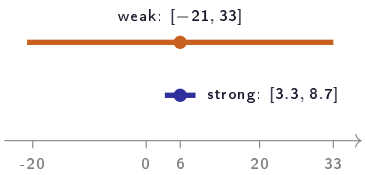
What happens when the offer barely changes attendance?

Dividing by a small number

Big Idea 7 — Weak instruments

Suppose the offer raises attendance by only 5 points (10% to 15%), and scores by 0.3. Then $\hat{\beta}_{IV} = 0.3/0.05 = 6$ again, but

$$SE \approx \frac{0.7}{0.05} = 14: \qquad 6 \pm 27.$$



A tiny first stage turns sampling noise in the ITT into a huge uncertainty about the effect.

Worse, with a weak instrument the 2SLS estimate is **biased towards the naive OLS** in finite samples, and the usual interval is too short.

Always report the first stage and its F-statistic.

The classical rule $F > 10$ (Staiger and Stock) is now considered lenient: valid 5% t-tests need a much larger F, or adjusted critical values (Lee et al., 2022).

Proof: where the weak-instrument bias comes from

Big Idea 7 — Weak instruments

MATHEMATICS

Sample version of the IV formula: $\hat{\beta}_{IV} = \beta + \dfrac{\widehat{\mathrm{Cov}}(Z, \varepsilon)}{\widehat{\mathrm{Cov}}(Z, T)}$.

- In the population $\mathrm{Cov}(Z, \varepsilon) = 0$; in a sample, $\widehat{\mathrm{Cov}}(Z, \varepsilon)$ is a small random number of order $1/\sqrt{n}$.
- Strong instrument: the denominator is large and stable, so $\hat{\beta}_{IV} \approx \beta$.
- Weak instrument: the denominator is close to zero and noisy, and correlated with the numerator: the part of T that chance links to Z carries motivation ε with it.

In the limit of an instrument with no first stage at all, $\hat{T}$ is pure noise correlated with ε in the same way as T, and $\hat{\beta}_{IV}$ tends to the biased OLS value. Approximately (Staiger and Stock, 1997):

$$\text{bias of 2SLS} \approx \frac{1}{F} \times \text{bias of OLS}$$

The clean case, again

Big Idea 7 — Weak instruments

So far the offer was drawn by lottery, it acted only through attendance, and nobody was discouraged by it. Many instruments used in practice are not lotteries: distance to a tutoring centre, a teacher's enthusiasm, a policy change.

What can go wrong with an instrument in practice?

Threats to an IV design

Big Idea 8 — An instrument can be relevant, while still invalid

The offer letter itself motivates students, even those who never attend	→ **exclusion violated**
Instrument = distance to a tutoring centre; families near centres differ (income, schools)	→ **independence violated**
Some students refuse because they feel labelled as weak	→ **defiers (monotonicity)**
The offer moves only 2% of students	→ **weak instrument**
Compliers are not the students the policy targets	→ **LATE not relevant**
Offered tutoring is 3 hours; outside tutoring is 20 hours	→ **different treatments**

An instrument solves hidden confounding only if its own path to the outcome is clean.

Proof: what an exclusion violation does

Big Idea 8 — An instrument can be relevant, while still invalid MATHEMATICS

Suppose receiving the offer letter raises **every** offered student's score by γ points directly, whether or not they attend (it encourages them to study). Redo the LATE proof:

always-takers and never-takers now contribute γ each; compliers contribute $Y(1) - Y(0) + \gamma$. Adding up:

$$E[Y \mid Z = 1] - E[Y \mid Z = 0] = P(\text{complier}) \cdot \text{LATE} + \gamma.$$

Dividing by the first stage $P(\text{complier})$:

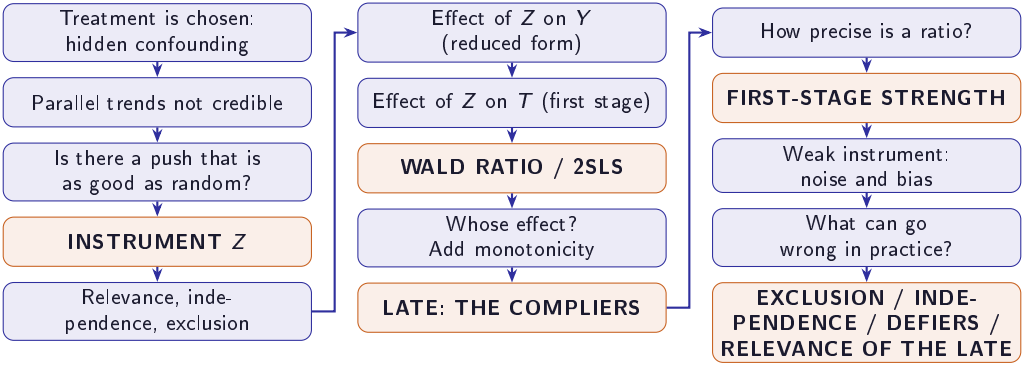
$$\hat{\beta}_{IV} \;\to\; \text{LATE} + \frac{\gamma}{P(\text{complier})}$$

Numbers. A letter effect of only $\gamma = 0.5$ point, with 50% compliers, adds $0.5/0.5 = 1$: the IV says 7 instead of 6. With 5% compliers the same small letter effect adds 10 points.

A weak first stage magnifies **every** small violation of exclusion or independence.

The complete logic of instrumental variables

Big Idea 9 — The complete logic of instrumental variables



The big message

Big Idea 9 — The complete logic of instrumental variables

An instrument recovers a causal effect from the part of treatment that chance decided.

Randomization made all of treatment random. DiD used time.

IV uses a **random push**: it does not need to control treatment, only something that moves it.

The price: the answer concerns the students the push moves (the compliers), and the exclusion restriction can never be fully verified.

But what if there is no instrument?

Big Idea 10 — But what if there is no instrument?

Everything in this chapter relied on finding a push that is as good as random and acts only through treatment.

Often there is none. But sometimes treatment is assigned by a **rule**:

"Tutoring is offered to every student who scores **below 50** on the autumn mock exam."

A student who scores 49 and one who scores 51 are almost identical, yet one is offered tutoring and the other is not. Near the cutoff, the rule behaves almost like a lottery.

a threshold that decides treatment.

Next chapter — Regression Discontinuity

Big Idea 10 — But what if there is no instrument?

So our next question becomes:

If treatment is decided by a cutoff, can comparing students just below and just above it give a credible counterfactual?

This question introduces the next chapter:

Regression Discontinuity Design (RDD)

For now, we stop there. The next chapter will explain **how and under what assumptions** a cutoff can be used for causal inference.

References: Imbens, G. W. and Angrist, J. D. (1994). Identification and estimation of local average treatment effects. *Econometrica* 62(2). Angrist, J. D., Imbens, G. W. and Rubin, D. B. (1996). Identification of causal effects using instrumental variables. *JASA* 91(434). Staiger, D. and Stock, J. H. (1997). Instrumental variables regression with weak instruments. *Econometrica* 65(3). Lee, D. S., McCrary, J., Moreira, M. J. and Porter, J. (2022). Valid t-ratio inference for IV. *American Economic Review* 112(10).